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Mixed L^2 basis set: STOs plus B-Splines. Calculation of the differential photoionization cross-section of Li_2

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Abstract

The convenience afforded by the use of L^2 functions in the calculation of the photoionization cross-section and of others properties depending upon the electronic continuum has prompted the examination of particular basis sets that, among those derived, could be able to mimic the continuum states in an extended region of space. Here results are reported for the differential photoionization cross-section of the Li_2 molecule, where a mixed L^2 basis set comprising multi-center STOs plus single-center radial B-Splines times spherical harmonics was employed.

1. Introduction

The use of L^2 basis sets for the computation of several properties inherent to the electronic continuum enjoys several advantages upon the more traditional methods [1–10] generally derived from those employed in the scattering calculations. In fact, the L^2 methods may rely upon the vast expertise gained in the quantum chemistry calculations for discrete states. Therefore, the computations may be performed at several and well understood levels of approximation and the multi-center character of the molecular target may be handled easily using multi-center basis functions.

The goal of selecting manageable bases which should describe satisfactorily both the multi-center nature of the neutral and ionic states of the molecules and the one-center character of the long-range behaviour of the electron continuum presents several difficulties. The pragmatic approach to employ a multi-center GTO basis plus a one-center basis comprising very diffuse Polynomial Spherical GTO (PSGTO), expressed as products of a Gaussian times a real spherical harmonic times a polynomial in r^2, has proved to be a workable one [11,12]. But still, in spite of the polynomial, needed to reduce the otherwise unmanageable problem of the redundancy of the one-electron basis, the description of the states in the continuum region may be sometimes unsatisfactory. The most obvious remedy would be to replace the diffuse set of PSGTOs by functions like B-Splines which would not yield severe redundancy problems. Unfortunately, we have found great difficulties to obtain the

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needed accuracy for all the integrals involved with GTO + B-Splines bases. Thus it was thought worthwhile to investigate the possibility of using a mixed basis set comprising a reduced basis of multi-center STOs

$$\chi_{n,\zeta_a,l,m}(|\mathbf{r}-\mathbf{r}_a|,\theta_a,\phi_a) = |\mathbf{r}-\mathbf{r}_a|^n e^{-\zeta_a|\mathbf{r}-\mathbf{r}_a|} \times S_{l,m}(\theta_a,\phi_a) \tag{1}$$

plus a much more extended basis of single center radial B-Splines times spherical harmonics

$$f_{n,l,m}(\mathbf{r}) = \frac{1}{r} B_{n,p}(r) S_{l,m}(\theta,\phi) \tag{2}$$

stretching up to rather large values of the radius. Hereafter, a stands for any relevant geometric center in the molecule, while $B_{n,p}(r)$ indicates a B-Spline of order p starting at the nth knot. With this mixed basis set of STOs plus B-Splines we were able to obtain a satisfactory accuracy for all the involved integrals.

This Letter reports the results obtained for the differential photoionization cross-section and the asymmetry parameter of the Li_2 molecule which, although very simple, may represent a good test due to the unusually long bond. As matter of fact it was employed [11] as a first application for the calculation of the photoionization cross-section with the GTO + PSGTO basis which, subsequently, were successfully used for the calculation of this same properties in several other molecules [13–15]. The Li_2 molecule has also been recently utilized to check a theoretical model for the evaluation of the triple differential cross-section for the $(e,2e)$ process [16]. Since experimental data are not yet available, the current results were compared with the previous GTO + PSGTO ones [11].

2. Theoretical method

In the dipole and vertical transition approximations for a sample of randomly oriented molecules, the differential cross-section for a transition from a initial state $|\Psi_0\rangle$ to a final state in the continuum $|\Psi^{(-)}_{j_\gamma,E,\hat{k}}\rangle$ due to the absorption of a single photon of energy ω may be expressed by the well known formula

$$\frac{d\sigma_{j_\gamma}(\omega)}{d\Omega_{\hat{k}}} = \frac{\sigma_{j_\gamma}(\omega)}{4\pi}\Big[1+\beta_{j_\gamma}(\omega)P_2(\cos\theta)\Big], \tag{3}$$

where $\sigma_{j_\gamma}(\omega)$ is the integral cross-section and $\beta_{j_\gamma}(\omega)$ is the asymmetry parameter. The index j_γ collects all the labels identifying the asymptotic ionic target state with $N-1$ electrons $|\Psi^{N-1}_{j_\gamma}\rangle$, spin coupled to the continuum orbital $\varphi_{j_\gamma,\epsilon,\hat{k}}$ describing the photoelectron wave at large r with energy $\epsilon = E - E^{N-1}_{j_\gamma}$ in the direction $\hat{k}$. The angle θ is measured between $\hat{k}$ and the photon polarization.

The quantities $\sigma_{j_\gamma}(\omega)$ and $\beta_{j_\gamma}(\omega)$ depend upon the transition matrix elements

$$\left\langle \Psi^{(-)}_{j_\gamma,E,l,m}|\hat{O}_\mu|\Psi_0\right\rangle \tag{4}$$

between the initial state and the final states in the continuum characterized by the asymptotic partial wave channel indices j_γ, l, m. The $\hat{O}_\mu$ indicates the transition operator, both the length and the velocity forms were considered, and μ stands for the photon polarization. The continuum states are determined through the projection of the reaction K-Matrix equations upon the grid of energy values yielded by a L^2 basis set calculation, where functions diffuse enough are included in order to account for a reasonable extended energy region. The calculations may be carried out at different level of approximation, from simple single-channel Static-Exchange Approximation (SEA) to more sophisticated Configuration Interaction (CI) type [17], as well as employing the transition densities as approximated by the RPA in the continuum. In last case, the method is called KM–RPA [18].

The technical details of the KM–RPA method are fully reported in previous publications [18,19], thus here we limit ourselves only to a few items specific to the particular mixed basis employed.

The KM–RPA approach relies upon a judicious selection of the orbitals used to build the multi-electron basis. In fact, in order to avoid the inclusion of extremely large values of l, required to account for the behaviour of the photoelectron wave in the region close to the nuclei, the method employs two different kind of multi-electron basis. The first one, called the *localized channel* (lc), includes determinants built with short range multi-center orbitals which will not contribute to the long range behaviour of the states in the continuum but are fundamental for their correct description close to the nuclei. The second one comprises a set of multi-electron basis built as anti-symmetrized product of some localized target ion states, say j, times partial waves expressed in terms of very diffuse single-center radial orbitals times spherical harmonics of assigned l, m values. Thus each set of these second type of multi-electron basis functions is characterized by the j, l, m indices and is called a *partial wave channel* (pwc).

To fit this framework, the complete collection of the basic excitation operators of the RPA themselves is divided into two sets. The first one may be referred to as $\mathsf{a}^{\dagger}_{j,\epsilon_v,\Gamma}\mathsf{a}_j$, where a_j annihilates a particle in the occupied SCF φ_j orbital while $\mathsf{a}^{\dagger}_{j,\epsilon_v,\Gamma}$ creates a particle in the $\varphi_{j,\epsilon_v,\Gamma}$ orbital projected upon the unoccupied SCF orbitals. The SCF orbital basis comprises basis functions, both STO and B-Splines, essentially of localized character. The index Γ refers to some irreducible representation of the molecular symmetry group and, as such, it may include several values l, m of the B-Spline basis functions. The $\varphi_{j,\epsilon_v,\Gamma}$s diagonalize the SEA Hamiltonian for the j hole. The second set includes excitation operators $\mathsf{a}^{\dagger}_{j,\epsilon,l,m}\mathsf{a}_j$, where the $\mathsf{a}^{\dagger}_{j,\epsilon_v,l,m}$ creates a particle in the very diffuse $\varphi_{j,\epsilon,l,m}$ orbital, projected upon the whole one-center radial B-Splines basis orthogonalized to the occupied SCF orbitals. The $\varphi_{j,\epsilon,l,m}$ diagonalize also the SEA Hamiltonian for the j hole. Thus the pwc 'unperturbed' asymptotic states are represented as $\mathsf{a}^{\dagger}_{j,\epsilon,l,m}\mathsf{a}_j|\Phi_{SCF}\rangle$. It becomes clear that this procedure yields different sets of states for each SEA Hamiltonian and these sets are not orthogonal amongst themselves.

It should be mentioned now that, since the K-Matrix technique involves delicate numerical interpolations and integrations into the continuum region, the lack of orthogonality between those sets may be the source for anomalies leading to spurious results. In the present instance the orbitals used in the lc functions as well as the occupied and unoccupied SCF-orbitals were built after projection upon a set of the multi-center STOs [Eq. (1)] plus the B-Splines [Eq. (2)] with several values of l, m extending to $r \leqslant R_0$ ($R_0 \sim 16$ a.u.). The whole B-Splines basis, reaching a rather large value of r ($\sim$50 a.u.) was utilized for each assigned l, m pairs to represent the partial waves of the pwcs. In order to overcome the danger indicated above, the lc set was then orthogonalized to the pwc ones, minimizing the redundancy of the whole basis set.

3. Basis set and integral evaluation

A rather extensive series of checks and trial calculations convinced us that for molecules involving only the atoms of the first row the multi-center STO basis may be restricted to a few 1s for each nucleus, since the single-center B-Spline basis with l up to 7–9 seems to be capable to account for the electronic distribution in the region not very close to the nuclei. Table 1 shows the excellent quality of the description achieved in the case of Li_2 molecule, here discussed, even compared to the traditional 6-311G GTO basis sets family [20]. The selection of the nodes depends obviously on the geometrical arrangement and the charges of the nuclei. In general, as expected, it was found profitable to thicken the knot distribution around the nuclei. The large-r distribution was taken to be uniform with internodes distances sufficient to insure a good representation of the waves whose k-vector was close to that of the most energetic photoelectron to be considered. A useful criterion to judge the quality of the node distribution for the localized region was to examine the energy spectrum of the one-electron Hamiltonian yielded by the resulting B-Splines basis. All in all the selection of the nodes, as well as the value of p, the order of the splines, was not a very critical point. The evaluation of the several types of integrals involving the 1s STOs and the

Table 1
Restricted Hartree–Fock total energies for Li_2 computed by standard GTO basis sets [20] and by the here described multi-center 1s STO + single-center B-Splines mixed basis set

Basis set type		E (a.u.)
Standard GTO basis sets	6-311G	−14.86980820
	6-311+G	−14.86981732
	6-311G*	−14.86998322
	6-311++G(2df)	−14.87000527
Double Zeta 1s STO + one-center B-Splines[a]	$l \leqslant 6$	−14.87112939
	$l \leqslant 9$	−14.87142977

[a] See text and Table 2 for details.

B-Splines $f_{n,l,m}$ was accomplished through several numerical integrations which, due to the piece-wise character of the B-Splines, turn out to be a very convenient procedure to obtain accurate values of the integrals. By exploiting the Laplace expansion and replacing the products of spherical harmonics with the same argument by the corresponding finite linear combinations, it is not difficult to find out that, in the simpler case of linear molecules, all the integrals reduce to radial integrations involving the following functions

$$F_{a,L,M}(r) = \int_{-1}^{1} \mathrm{d}\eta\, \mathrm{e}^{-\zeta_a\left(r^2+z_a^2-2rz_a\eta\right)^{1/2}} P_{L,M}(\eta), \tag{5}$$

$$F_{a,b,L,M}(r) = \int_{-1}^{1} \mathrm{d}\eta\, \mathrm{e}^{-\zeta_a\left(r^2+z_a^2-2rz_a\eta\right)^{1/2}-\zeta_b\left(r^2+z_b^2-2rz_b\eta\right)^{1/2}} P_{L,M}(\eta) \tag{6}$$

times some power of r depending upon the $r_<^L/r_>^{L+1}$ term of the Laplace expansion and upon the spline polynomials. z_a and z_b are the coordinates of the nuclei a and b in the linear molecule placed along the z-axis. By several tests it was found that a self-adaptive technique of the one-dimensional quadrature was the most efficient way to obtain satisfactory accuracy for the $F_{a,L,M}(r)$ and $F_{a,b,L,M}(r)$ radial functions.

Almost all the integrals are expressible as a finite sum of multipole contributions. The two types of two-electron integrals which should be computed by an infinite sum of such contribution are $[1s_a, f_{n_1,l_1,m_1}|1s_b, f_{n_2,l_2,m_2}]$ and $[1s_a, 1s_b|1s_c, f_{n,l,m}]$, the last one in the general case where functions $1s_a$ and $1s_b$ belong to different centers. These integrals, applying the procedure sketched above, reduce to the following radial integrations

$$\begin{aligned}
&[1s_a, f_{n_1,l_1,m_1}|1s_b, f_{n_2,l_2,m_2}] \\
&\quad\propto \sum_L \sum_{\lambda_1} \sum_{\lambda_2} \sum_M \frac{[(2\lambda_1+1)(2\lambda_2+1)]^{1/2}}{2L+1} \\
&\quad\times \langle S_{\lambda_1,0}|S_{l_1,m_1}S_{L,M}\rangle \langle S_{l_2,m_2}S_{L,M}|S_{\lambda_2,0}\rangle \int_0^\infty r_1\,\mathrm{d}r_1 \\
&\quad\times \int_0^\infty r_2\,\mathrm{d}r_2 B_{n_1,p}(r_1) \\
&\quad\times F_{a,\lambda_1,0}(r_1)\frac{r_<^L}{r_>^{L+1}} B_{n_2,p}(r_2) F_{b,\lambda_2,0}(r_2),
\end{aligned} \tag{7}$$

$$\begin{aligned}
&[1s_a, 1s_b|1s_c, f_{n,l,m}] \\
&\quad\propto \delta_{m,0} \sum_L \sum_\lambda \left(\frac{2\lambda+1}{2L+1}\right)^{1/2} \langle S_{\lambda,0}|S_{L,0}S_{l,m}\rangle \int_0^\infty r_1^2\,\mathrm{d}r_1 \\
&\quad\times \int_0^\infty r_2\,\mathrm{d}r_2 F_{a,b,L,0}(r_1)\frac{r_<^L}{r_>^{L+1}} F_{c,\lambda,0}(r_2) B_{n,p}(r_2).
\end{aligned} \tag{8}$$

The radial integrations were carried out using Legendre Gaussian and Laguerre Gaussian points. It was convenient to prepare tables of the $F_{a,L,M}$ and $F_{a,b,L,M}$ functions evaluated for a grid of radial points. The grid was selected by using from 10 to 20 Legendre Gaussian points, depending from the extension of the interval, between each couple of adjacent nodes, and around 6–10 points between each couple of the previous Gaussian points.

The matrix elements $\langle S_{\lambda,0}|S_{L,0}S_{l,m}\rangle, \langle S_{\lambda_1,0}|S_{l_1,m_1}S_{L,M}\rangle$, etc. between real spherical harmonics impose, besides some trivial constraints upon the ms and M, also a triangular condition $\Delta(\lambda, L, l)$ upon the allowed values of λ and L. In the case of Eqs. (7) and (8), the sum of the multipoles was carried out up to a maximum $L = 30$, which guaranteed the convergence to the accuracy reported below. The maximum values employed for the λs were those compatible with $\Delta(\lambda, L, l)$.

On the contrary, if $1s_a 1s_b$ is a single-center distribution upon z_a, an easy shortcut may be found to speed calculations and increase accuracy:

$$\begin{aligned}
&[1s_a, 1s_b|1s_c, f_{n,l,m}] \\
&\quad\propto \delta_{m,0}(2l+1)^{1/2} \int_0^\infty r\,\mathrm{d}r B_{n,p}(r) \\
&\quad\times \int_{-1}^{1} \mathrm{d}\eta V\left[\left(r^2+z_a^2-2r_a z_a\eta\right)^{1/2}\right]. \\
&\quad\times \mathrm{e}^{-\zeta_d\left(r^2+z_d^2-2rz_d\eta\right)^{1/2}} P_{L,0}(\eta)
\end{aligned} \tag{9}$$

with

$$V(\rho) = \frac{1}{(\zeta_a+\zeta_b)^2}\left\{\frac{2}{\zeta_a+\zeta_b}\,\frac{1-\mathrm{e}^{-(\zeta_a+\zeta_b)\rho}}{\rho} - \mathrm{e}^{-(\zeta_a+\zeta_b)\rho}\right\}.$$

The integrals obtained by our code employing, as said, Legendre and Laguerre Gaussian techniques for radial quadrature, have been compared with the corresponding ones calculated by resorting to a completely self-adaptive quadrature with very stringent accuracy requirements. In any case the relative accuracy attained, never inferior to 10^{-10}, may be considered satisfactory and sufficient to not cause any numerical problem in the KM–RPA procedure. A check on the convergence for some of the most critical cases is graphically shown in Fig. 1.

It turned out that the computer code, although not completely optimized, is able to evaluate an average of 3–4 $\times 10^4$ two-electron mixed-basis integrals per second. [1]

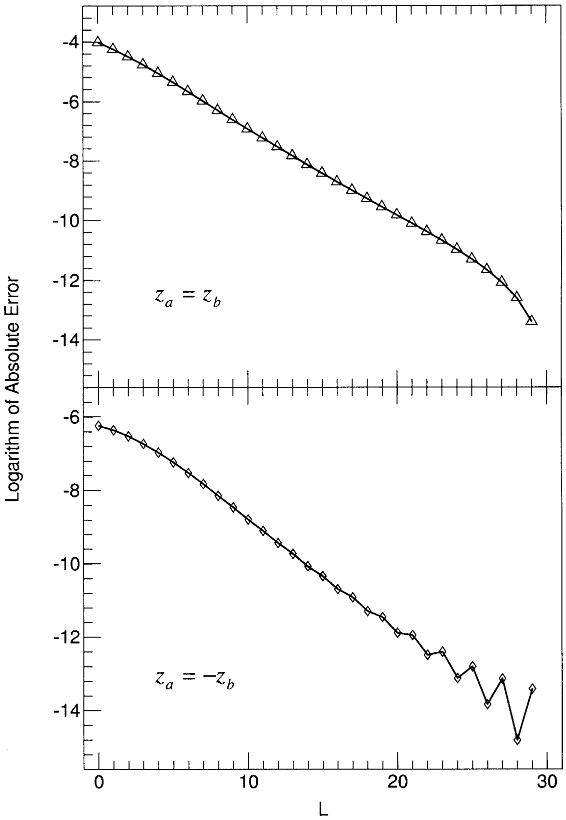

Fig. 1. Behaviour of the convergence of the Laplace expansion for two-electron integrals $[1s_a\, f_{n,0,0}|1s_b\, f_{n,0,0}]$ [see Eq. (7)]. The logarithm of the absolute error is plotted as a function of the maximum L employed. Smooth convergence shown in the graph on top arises when the 1s, here both with $\zeta = 4.3069$, belong to the same nucleus ($z_a \equiv z_b$), whereas an oscillatory trend takes place when they are placed on opposite nuclei ($z_a = -z_b$). The s-kind B-Spline involved $f_{n,0,0}$ is the $n = 15$ one, mostly localized around both nuclei. The final values ($L = 30$) are $2.214542862839 \times 10^{-2}$ ($z_a \equiv z_b$) and $2.0940845068741 \times 10^{-3}$ ($z_a = -z_b$).

4. Results

In order to obtain a significant benchmark for this new mixed basis set and for our code, all the calculations were performed in a way to be as close as possible to the previous ones employing the GTO + PSGTO basis set [11]. Therefore, any parameter like bond length, experimental ionization potential and photon energy range were identical to those already used. Since the one-center B-Splines are responsible also for a good description of the localized charge distribution, it was thought necessary to investigate the l-convergence. Here, after using the old $l = 7$ (together with short-range B-Splines for SCF orbitals up to $l = 6$), we performed another set of calculation pushing l up to 9 for both SCF and waves functions. These calculations were used to check if an acceptable convergence was already achieved by the first basis set.

The simple photoionization process we took into account involved the absorption of a single photon with energy up to 50 eV by a Li_2 molecule in its ground state. In the vertical transition and dipole approximations, because of the $D_{\infty h}$ molecular symmetry, the single pathways allowed to the photoelectron to go from the valence shell of the ground state ${}^1\Sigma_g^+$ into a wave leaving the parent ion in its ground state ${}^2\Sigma_g^+$ are

$$\begin{aligned} &\epsilon\sigma_u({}^1\Sigma_u^+) \leftarrow \sigma_g \quad \|\text{-bond polarized photon,} \\ &\epsilon\pi_u({}^1\Pi_u) \leftarrow 2\sigma_g \quad \perp\text{-bond polarized photon.} \end{aligned} \tag{10}$$

The functions required to obtain the irreducible representations belonging to this point group are easily built from single center B-Splines [Eq. (2)] placed in the middle of the Li–Li bond. Therefore, according to symmetry, the only pwcs involved here are those labeled by $(l, 0)$ for σ_u and $(l, \pm 1)$ for π_u for the odd values of l.

[1] Estimate on a Pentium III 1.7 GHz machine.

A summary of the composition of the basis set is reported in Table 2. The STO set centered upon each Li atom, used in SCF for description of core, was chosen to be an optimized Double Zeta 1s [21]. Every (l, m) subset was built from fifth-order B-Splines computed upon a unique knot sequence and the number of functions retained in the calculations was chosen in dependence of the localized or diffuse character of the electronic states involved. In fact, in the former case, which involves molecular orbitals (SCF) and lc states, 32 B-Splines were employed to cover the space up to 16 a.u. about, a value enough to ensure the right description of localized orbitals as checked by several trial calculations. In the latter case, an accurate representation of the continuum waves required 67 functions extending to 50 a.u.

The ground state was determined assuming an equilibrium bond length of 2.6725 Å (5.05029 a.u.) [11,22]. The basis set comprising B-Splines up to $l = 6$ gave $-14.871\,183$ a.u. as SCF energy and 4.949 eV as Koopman ionization potential, to compare with the experimental value of 5.0 eV [22]. As shown in Table 1, the total SCF energy did not appreciably change after the inclusion of functions with l up to 9 into the basis set, proving that the convergence is quickly achieved in the case of occupied molecular orbitals. An estimate of the quality of different basis sets for SCF may be made from the data reported in Table 3.

Table 2
Detailed description of the mixed 1s STO + B-Splines basis set employed in this work. For each subset is also indicated the part of calculations where it was involved. Atomic units are understood where needed

	r_0	ζ		Calculations
Double Zeta 1s STO[a]	2.52515	4.3069 2.4573		SCF, lc
	r_0	# Used	R_{box}	Calculations
B-Splines in (l, m) subsets, order $p = 5$[b]	0.0	2–33	~16	SCF, lc
		2–68	50	pwc

[a] Eq. (1) now reduces to $\chi_a(\mathbf{r}) = e^{-\zeta_a|\mathbf{r}-\mathbf{r}_a|}$.
[b] See Eq. (2).

Table 3
Koopman ionization potentials $(-\epsilon)$ and total energies E from SCF calculations on Li_2 $^1\Sigma_g^+$ ground state performed with standard GTO and the mixed basis sets described in the text

Orbital I.P. (eV)	6-311G* Ref. [11]	DZ 1s STO + B-Splines $l \leqslant 6$	DZ 1s STO + B-Splines $l \leqslant 9$
$1\sigma_g$	66.736	66.755	66.752
$1\sigma_u$	66.727	66.746	66.743
$2\sigma_g$	4.947[a]	4.949	4.950
E (a.u.)	−14.869986	−14.871183	−14.871419

[a] The experimental value is 5.0 eV [22].

Due to their great flexibility, B-Spline functions yield a L^2 representation of electronic continuum extremely more detailed than any other set already

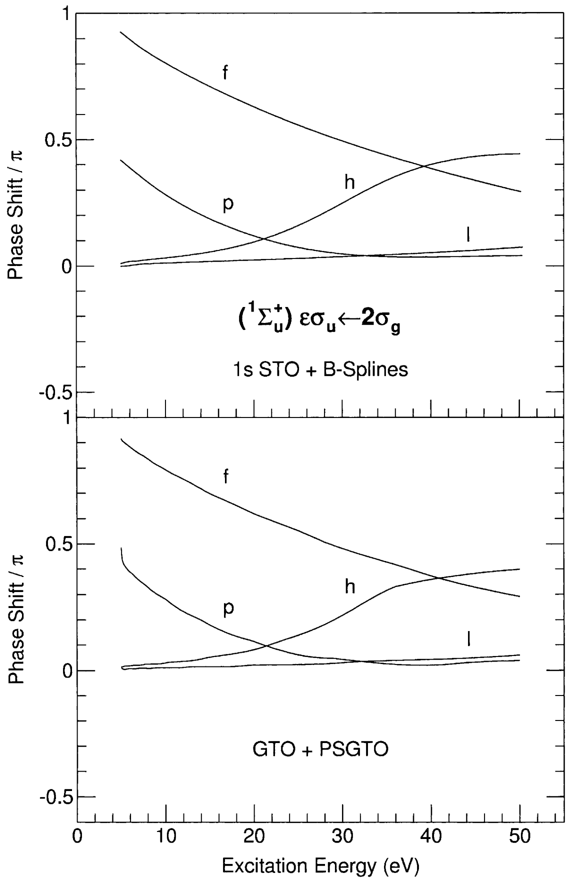

Fig. 2. Phase shifts of the pwc orbitals with respect to pure Coulomb waves with same energy for the $^1\Sigma_u^+$ continuum symmetry of Li_2 $(2\sigma_g)^{-1}$. Current results of 1s STO + B-Splines calculations are shown in upper figure, earlier GTO + PSGTO results [11] in lower figure.

used, e.g., PSGTO, filling the energy spectrum with more numerous wave packets. This allows one to obtain an easier and more accurate determination of the phase shifts belonging to pwc orbitals by the usual interpolation techniques [23], without resort to the complicate procedure needed sometimes with the Gaussian basis sets [11]. The phase shifts of pwc orbitals up to $l = 7$, compared to those computed by GTO + PSGTO basis set and the Separate Channel K-Matrix (SC-KM) technique [24], are reported in Figs. 2 and 3 for the $^1\Sigma_u^+$ and the $^1\Pi_u$ states, respectively. The behaviour is quite similar in both cases. It may be observed that the phase-shifts belonging to the $^1\Pi_u$ partial waves (Fig. 3) are much smaller than those of $^1\Sigma_u^+$ (Fig. 2), due to the reduced value of the π orbitals in the region of the largest variation of the Coulomb potential. The small differences between the two sets of calculations validate, in our opinion, the previous approach in such simple cases.

The calculation of the photoionization cross-section and the asymmetry parameter was carried out at RPA level in both length and velocity gauges and the results are shown in Fig. 4. A comparison between these new results and the older ones is then reported in Fig. 5, where the mixed gauge values are shown. In a more concrete way, considering the mixed gauge, the integral

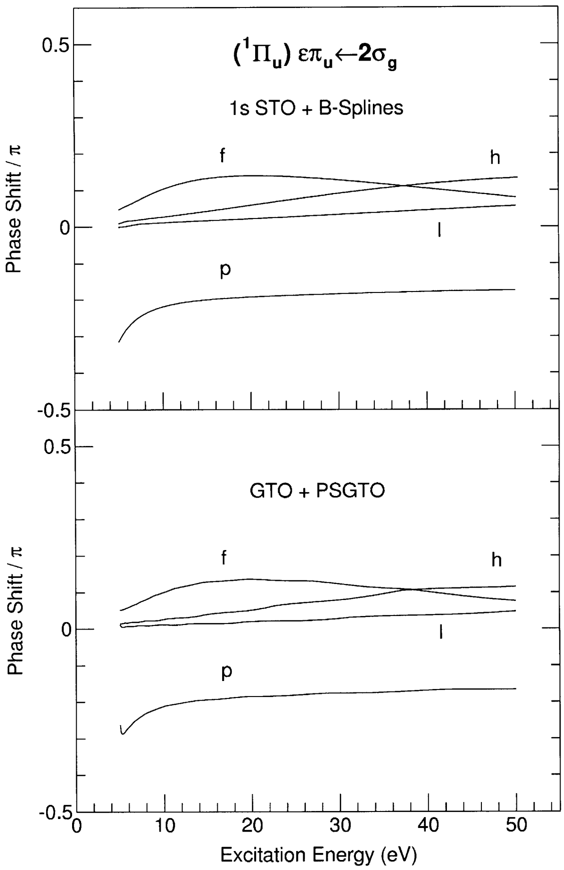

Fig. 3. Phase shifts of the pwc orbitals with respect to pure Coulomb waves with same energy for the $^1\Pi_u$ continuum symmetry of Li_2 $(2\sigma_g)^{-1}$. Current results of 1*s* STO + B-Splines calculations are shown in upper figure, earlier GTO + PSGTO results [11] in lower figure.

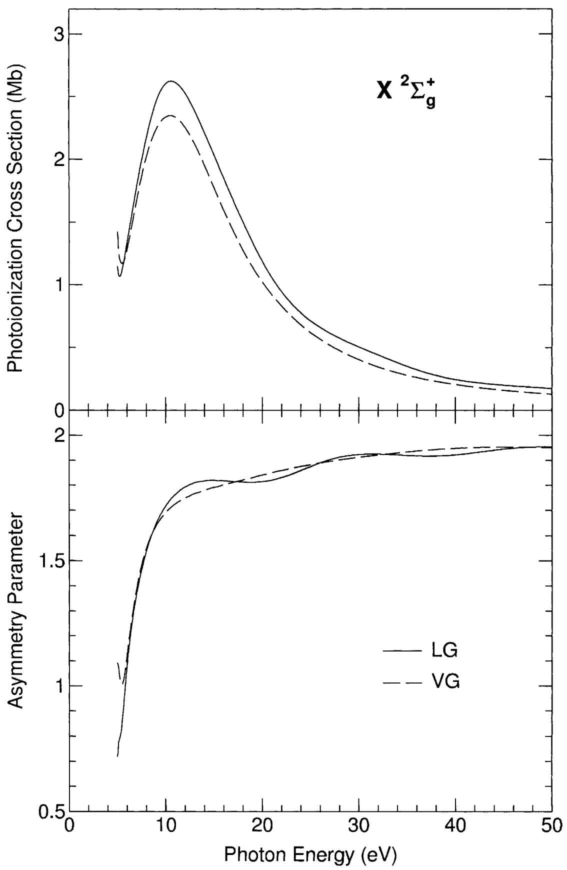

Fig. 4. Partial cross-section and asymmetry parameter for the photoionization of Li_2 leading to the $X^2\Sigma_g^+$ ion state, in length (LG) and velocity (VG) gauges, computed by the mixed 1*s* STO + B-Spline basis set. Localized channels and pwcs with quantum number *l* up to 7 were considered.

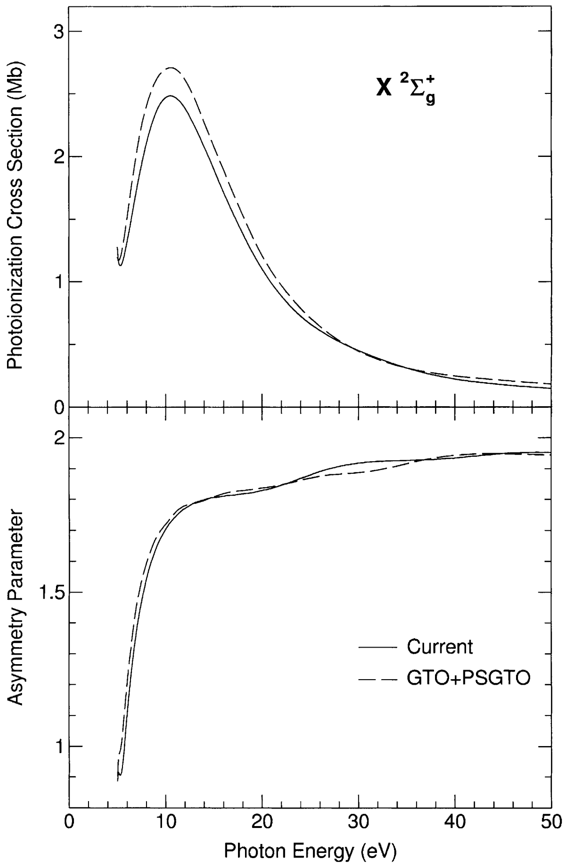

Fig. 5. Comparison of Li_2 $X^2\Sigma_g^+$ photoionization cross-sections (upper graph) and asymmetry parameters (lower graph) in mixed gauge computed by the new 1*s* STO + B-Splines mixed basis set (full line) and by the GTO + PSGTO basis (dashed line) [11]. Localized channels and pwcs with quantum number *l* up to 7 were included in both calculations.

cross-sections differ by 9% on average, and the asymmetry parameters less than 2%. It may be noted that the data reported in Fig. 4 display a slight disagreement between the two gauges utilized, contrary to the RPA properties. This is quite probably due to the neglect of the $1\sigma_g$ and $1\sigma_u$ excitations more than to an inadequacy of the basis set.

The calculations performed including in both SCF and pwc basis set B-Spline functions up to $l = 9$ show, always in mixed gauge, an average increase of cross-section values limited to 3% with respect to $l = 7$, whereas the differences in the asymmetry parameter are negligible (~0.3%). This definitely suggests that, even for B-Splines, the contribution from spherical harmonics beyond $l = 7$ may be neglected with respect to the considered observables, as already found out with PSGTOs [11].

5. Conclusions

We performed K-Matrix RPA calculations for the single-photon photoionization of Li_2 molecule in the fixed-nuclei approximation with the purpose of checking the abilities of a new mixed basis set, built by multi-center STO functions and single-center radial B-Splines times spherical harmonics, here proposed, to describe accurately the electronic continuum. The results reported show that this basis set is able to attain an excellent accuracy of the matrix elements involved into the calculations and insure consequently, together with an improved description of the electronic waves, a reliable account of the quantities related to the photoionization processes. It appears, therefore, that the present approach may be employed as a valid tools for a more accurate evaluation of the host of properties depending on the electronic continuum.

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